

KTV Helper: Smartwear Companion

Wrist-Based Interaction Design for Real-Time Service Requests, Physiological Safeguards, and Split-Billing

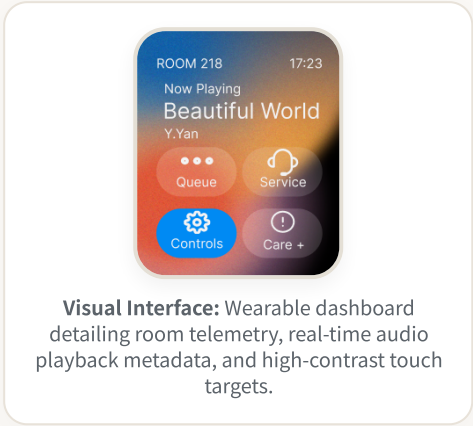
Course: IDEA9105 Interaction Design
Project Code: 9105-SmartwearCompanion

Student Name: |
Student ID:

01 DESIGN SOLUTION

A wrist-worn smartwatch companion designed to mitigate the auditory, cognitive, and physiological barriers inherent to high-decibel KTV social environments.

By offloading high-friction group configuration tasks from smartphones to a tactile, glanceable wearable interface, the system prioritizes immediate, context-aware micro-interactions.



Visual Interface: Wearable dashboard detailing room telemetry, real-time audio playback metadata, and high-contrast touch targets.

TARGET ENVIRONMENT CONSTRAINTS:

Auditory overload (ambient noise >85dB), extreme low-light, alcohol-induced cognitive and psychomotor degradation, and high-friction transportation/payment logistics upon event conclusion.

02 DEEPENED DESIGN PAIN POINTS

Multi-dimensional analysis demonstrating the failure modes of standard smartphone interfaces within high-decibel, high-energy social contexts.

AUDITORY MASKING & SPATIAL ISOLATION

High-decibel environments (85–95 dB) render voice-based interaction ineffective and mask standard auditory notifications, necessitating alternative feedback channels.

COGNITIVE & PSYCHOMOTOR IMPAIRMENT

Intoxication degrades fine-motor control and executive function, making complex menu hierarchies and precise alphanumeric input high-friction and error-prone.

UNDETECTED PHYSIOLOGICAL STRAIN

The combination of high-exertion vocal performance and alcohol consumption causes acute cardiovascular spikes, which go undetected by the user without continuous passive monitoring.

GROUP COORDINATION FRICTION

Managing service requests and dividing bills during late-stage social events introduces significant cognitive load, resulting in social friction and payment delays.

03 ITERATIONS & REFLECTIONS

Iterative refinement of watch face spatial layouts and interaction models across three design cycles.

●

Iterative Grid Paradigm

Usability

Replaced standard list structures with high-contrast, large-target touch bubbles in a 2x2 grid to leverage spatial memory and accommodate degraded motor precision.

●

Intentional Friction States

Efficiency

Introduced distinct intermediate confirmation screens (e.g., 'Calling...' with countdown) to prevent accidental service invocations from erratic tactile inputs.

●

Zero-Input Dispatch

Safety

Eliminated alphanumeric input by utilizing pre-configured home coordinates, providing immediate, glanceable vehicle confirmation (license plate and color).

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Visual Design Consistency

Adopts a deep charcoal canvas background to support OLED energy conservation, paired with rich sunset red-orange gradients (`#E0533C` to `#11161B`) for warm room visuals and vibrant active blue triggers.

📊

Usability Evaluation Success (SUS)

System Usability Scale (SUS) scores showed a significant increase from an initial **62.4** (Round 1) to a final **85.8** (Round 3), demonstrating substantial improvement in task completion rate.

04 KEY VISUAL CHECKPOINTS

1

Tactile Service Request

Issues discrete service calls to KTV staff, confirming success via a localized haptic vibration pattern.

2

Physiological Safety Net (Care+)

Continuously monitors PPG heart rate trends, automatically triggering localized prompts or peer alerts during abnormal cardiovascular spikes.

3

Zero-Setup Transit Dispatch

Initiates automated ride-hailing to the user's primary address with a single tap, displaying essential driver credentials.

4

Glanceable Billing Split

Computes and displays individual cost divisions directly on the wrist, synchronizing status across group devices to minimize transaction friction.

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Sensor & Physical Integrations

Uses built-in **Optical PPG sensor** for heart rate tracking, **linear resonant actuator (LRA)** for haptic prompts, and **GPS** for safe dispatch.

05 PRIMARY WIREFLOW JOURNEY

Sequential task flows tracing cross-page transition states: home dashboard navigation, service classification menu, optical PPG vitals monitoring, hold-to-confirm staff calls, GPS safe transit dispatch, and peer-synchronized cost settlement.

A horizontal sequence of six watch face screenshots connected by right-pointing arrows.
1. **Home View**: Shows room status, song, and service buttons. Caption: 'Active room & song status'.
2. **Service List**: Shows a menu of services like Heart Rate, Call for Water, One-Tap Ride, and Care+. Caption: 'Select service category'.
3. **Vitals Check**: Shows a heart rate monitor with '98 bpm' and 'Good' status. Caption: 'Live heart rate sensor tracking'.
4. **Water Calling**: Shows a 'Calling' button with a glass icon. Caption: '2-second hold cancellation state'.
5. **Ride Active**: Shows a 'One-Tap Ride' button with a car icon and 'Ride Arriving Plate No. 12345'. Caption: 'Taxi incoming plate code tracking'.
6. **Split Payment**: Shows a bill summary with 'Total Amount \$248.00' and 'Split Method Equal'. Caption: 'Instant equal bill division summary'.

KTV Helper Smartwear Companion Poster | Exhibition Layout

Page 1 of 2 | Supporting Evidence Level: High

KTV Helper: Smartwear Companion Appendix

Supporting Evidence: High-Fidelity Interface Sets, Hand-Drawn Sketches, and Academic References

Course: **IDEA9105 Interaction Design**
Evidence Level: **Comprehensive Academic Submission**
Document Code: **IDEA9105-A4-SMARTWEAR-APPENDIX**

A FULL 10-SCREEN SMARTWEAR INTERFACE SET

High-fidelity screen displays mapping the complete smartwatch companion subsystem.

1. Home View
Room, track, shortcuts.

2. Service List
Health, Service, Ride options.

3. Heart Rate
PPG sensor display.

4. Call Water 1
Intermediate calling state.

5. Water Request
Staff confirmation alert.

6. Hail Ride 1
Taxi booking loader.

7. Ride Arrived
Driver plate code.

8. View Bill
Total room fees list.

9. Settle Split
Equal split payment.

10. Settings
Vibration & toggles.

B HAND-DRAWN SKETCHES & LOW-FIDELITY WIREFLOWS

Evidence of visual ideation, paper wireframes, and digital wireflows demonstrating incremental iterations.

B.1 Ideation Sketches (Week 8)
Pencil concepts mapping round watch target sizing and circular button bubble trials.

B.2 Paper Wireframes (Week 9)
Sketched rectangular casing dimensions, cancel delay rings, and swipe target outlines.

B.3 Digital Lo-Fi Flow (Week 10)
Figma grey-box interactive wireflow tracing the primary watch assistant journey paths.

C USABILITY REVIEW & HEURISTIC REFLECTIONS

HEURISTIC CATEGORY	IDENTIFIED WATCH FRICTION POINT	DESIGN CORRECTION & WEARABLE ADAPTATION
Visibility of System Status	Absence of audible feedback prevents confirmation of service requests in high-decibel (85+ dB) social spaces.	Designed prominent full-screen state transitions paired with a distinct, sustained haptic vibration pattern.
Error Prevention	Intoxication impairs psychomotor skills, causing accidental touch inputs on adjacent control elements.	Integrated a 2-second touch-and-hold delay timer for high-consequence operations (Safe Ride, Care+ trigger) to permit cancellation.
Match System & Real World	Standard mobile transaction terminology introduces excessive cognitive load under fatigue.	Adopted natural, task-oriented phrasing ("Total Amount", "Your Share", "Split Summary") paired with visual avatars.
Flexibility & Efficiency	Navigating backward from deep menu levels requires repetitive, precise swipe gestures.	Provided a persistent, high-contrast back button and mapped physical crown clicks for immediate home navigation.

CONSULTATION MEETING LOGS

Week 8 Review: Creative Director recommended moving away from complex mobile-style inputs. Instructed focusing on wearable-specific constraints: zero text inputs, high contrast, and haptic indicators.

Week 11 Evaluation: Peer testing showed

INTERACTIVE PROTOTYPES

Access the high-fidelity interactive companion watch prototype via the verified path below:

<https://www.figma.com/proto/>

ACADEMIC REFERENCES

Batchu, V. (2020). Designing better Apple Watch apps. *UX Collective*. <https://uxdesign.cc/designing-wearable-apps-ef941d11a16f>

Brewster, S. A., Lumsden, J., Bell, M., Hall, M., & Tasker, S. (2003). Multimodal 'strong' haptics for mobile devices. In *ACM CHI*, 173-180.

Nielsen, J. (1994). *Usability Inspection Methods*. John Wiley & Sons.